

Pricing invoices against a number nobody can see

How we turned a 60-second pricing game into a measured payoff function, a market memory, and a learning loop, and what the data taught us on the way.

1st

×3 in v2's first six rounds

0.08

vs 0.41 · memory vs LLM log error

5 to 16 s

per round, one call

What is hard about pricing a claim from both sides

As the handyman (issuer)

- You never see t . Charge a cent over it and the charge flips from "paid by all 16" to "paid by the lenient few".
- The fair value depends on a policy (sub-limits, declared values, exclusions), a market rate, and an invoiced quantity. Three sources of error that compound.
- One or two items per round are worth more than all the others together. Being 10 % timid on a whale costs more than being wrong on ten call-out fees.

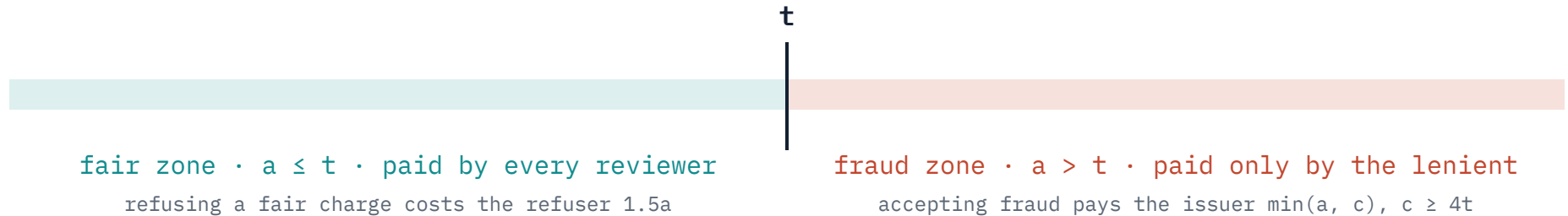
As the insurer (reviewer)

- Refusing a fair charge costs 1.5x; accepting a fraudulent one costs the full amount. The mistakes are asymmetric and you cannot see which one you are making.
- Coverage is binary and policy-specific ($t = 0$ when not covered): the same line item is payable in one case and excluded in the next.
- 16 opponents bill you every item. Their charges are a signal about t that you only get to read after the round.

And for both: 60 seconds, from key release to board close

Decrypt, read a policy of up to 65k characters and an invoice of up to 39 lines, decide a and b for every line, submit. A model call that takes 33 s against that clock is a missed round. A missed round refuses every fair charge at 1.5x while charging nothing: the single most expensive outcome in the game.

The payoff, made explicit: everything hinges on where a sits relative to t



Issuer: the payoff is a step

$$R(a) = a \cdot P(t \geq a) + 0.5 \cdot t \cdot P(t < a)$$

Measured on the field: revenue per opponent above t is flat at $0.48 \cdot t$ to $0.53 \cdot t$ for every overcharge ratio from 1.0 to 5. Crossing t forfeits half at once, and nothing more is gained beyond it.

Reviewer: accept iff $P(\text{fair}) > \frac{2}{3}$

accept: cost a · refuse: cost $1.5a \cdot P(\text{fair})$
 $\Rightarrow b = Q(\frac{2}{3})$ of an honest belief

With coverage inside it: $b = \sup\{a : P(\text{cov}) \cdot S(a) \geq \frac{2}{3}\}$. Everything hinges on the belief being honest, not on the quantile being tuned.

The strategy we went in with

Let the model read, let the math decide

- One LLM call reads policy, description and invoice verbatim and returns, per line: covered?, related?, t_{low} / t_{mid} / t_{high} .
- A lognormal belief around t_{mid} . The charge maximises mean minus 0.55 sd and never exceeds the median; the limit is the 1/3 quantile.
- A fast safety pass submits first and the full pass overwrites it. Later: long invoices are split into parallel chunks so one slow call cannot cost the round.

Learn from the leaderboard after every round

- The public payout feed proves a bracket $[t_{\text{lo}}, t_{\text{hi}})$ per item: a charge paid by everyone is fair, a charge refused by everyone and paid 0 is fraud.
- Fit a bias and a sigma between our t_{mid} and the proven brackets, fit how often reviewers accept over-charges, and re-price every stored round to test a change in seconds.
- Decide changes on money, expected and pessimistic, never on a single metric.

What the rounds taught us, and what we changed each time

WHEN	WHAT HAPPENED	WHAT WE CHANGED
rounds 10, 15	one long model call missed the 60 s deadline; the whole round shipped with fast-pass prices	split the full pass into parallel chunks of at most 10 items; submit after every chunk
round 27	the invoice parser met an unknown unit ("68 lines") and threw the entire invoice away	layout-only POS and label parser with no unit vocabulary; the invoice text stays authoritative
rounds 12 to 30	calibration pinned at its clamps; material, labour and diagnostics under-priced by 2x	wider clamp, per-category bias, then a size-dependent spline (one slope below a pivot, one above, a whale segment)
rounds 30 to 44	minus 165k: under-charging (median a/t_lo 0.60 to 0.79) and two blank rounds; the fitted acceptance curve was a censoring artifact	step-EV charge on the measured payoff, $b = Q(1/3)$ derived, pure EV, whale b-floor: 31 % less re-priced regret
rounds 31, 34	nobody pressed the button in time; a Windows path and 7z chain broke the key fetch	pure-Python key fetch and extract; autoplay launches every round 90 s early
research pass	memory of past proven brackets predicts recurring items 4x better than the model; whales decide rounds	engine v2 : memory prior, short prompt, p_covered, 3 samples, one call in 5 to 16 s
round 54	a past coverage refusal became a price anchor of EUR 47 to 77 for a EUR 2,000 unit: minus 48.6k	v4 : memory carries price from paid observations only; refusals never lower coverage

Six things the data told us that we did not expect

63

rd pct

t is exogenous. It sits at the 63rd percentile of the field's charges (IQR 0.33 to 0.86). There is a real number to estimate. The market is not the answer key.

0.08

vs 0.41

Memory beats the model. For recurring line items, past proven brackets predict t with MAD 0.08 to 0.10 in log space; the LLM 0.36 to 0.41 on the same items. Averaging the two is *worse* than memory alone.

-0.03

The LLM's interval is uninformative. Its stated q10 to q90 has correlation -0.03 with its realised error. σ must come from measured residuals.

0.5 t

Overcharging is flat. The fitted $p_0 \cdot r^{-k}$ curve ($k < 1$) claimed revenue rises forever. That was a censoring artifact. Refusals reconstructed, it is a flat plateau.

2

items

Whales decide rounds. The best team's entire lead was two items charged at $\sim 2\times$ the field median and still fair (9,585 vs our 4,771; 8,942 vs 5,285).

-116k

Presence beats pricing. Two rounds with no board cost more than all pricing errors of the same 15-round window combined.

One pipeline, two engines, a loop that learns after every round

Extract

decrypt, pypdf, POS +
label parser (layout-only,
no unit vocabulary)

~0.3 s

Estimate

one OpenAI call; v1
chunks ≥ 10 items in
parallel; v2 three
samples, per-item
median

v1 20 to 40 s, v2 5
to 16 s

Price

lognormal belief, then
step-EV charge, $\frac{2}{3}$ -rule
limit, whale floors,
coverage inside the rules

ms

Submit

first board the moment it
exists; later passes
overwrite (last write wins)

first board ~15 s

Learn

truth brackets from public
payouts; calibration refit;
reliability alarm; market
memory

after close

Engine v1: the model prices everything

$t_{\text{low}}/t_{\text{mid}}/t_{\text{high}}$ from the prompt; a 3-segment log-log spline corrects size-dependent bias (β below a pivot, above it, and a whale segment); σ by interval-censored MLE; per-category bias buckets with shrinkage.

Engine v2: memory first, model second

Per-line memory prior from every past proven bracket, precision-weighted against the LLM; a 1,400-character prompt with anchors inline and `p_covered` instead of a boolean; autoplay launches every round 90 s early.

Autoplay: reads the schedule, never re-plays a logged round, never runs two rounds at once. Being present is the whole job.

The belief is the whole decision. a and b are derived, not tuned

```
ln t | covered ~ N(μ, σ²)
μ = precision-weighted( memory μ_M/σ_M² , LLM ln q50/σ_L² )
a = argmax_a P(cov) · [ a · S(a) + 0.5 · E[t ; t < a] ] (capped at Q(0.55))
b = exp( μ + σ · Φ⁻¹(1 - ⅓ / P(cov)) )
whale (q50 ≥ 2k): b ≥ posterior median
```

- **Why pure EV, not mean – λ·sd:** a round is a portfolio of 10 to 20 items × 16 opponents; per-item shading forfeited 21 to 40 % of provably collectable income. Re-priced regret fell 748,539 to 516,256 (–31 %).
- **Why B_QUANTILE = ⅓ is derived:** the tuned values (0.33 to 0.40) were patches for a mis-centred belief. Centre μ instead (live_shift, reliability-monitored) and the quantile is the number the math gives.
- **Why memory is price, not coverage:** a past refusal predicts the next case's coverage only 40 % of the time. Policies vary per case; market prices do not.

Two-sided counterfactual (the honest test)

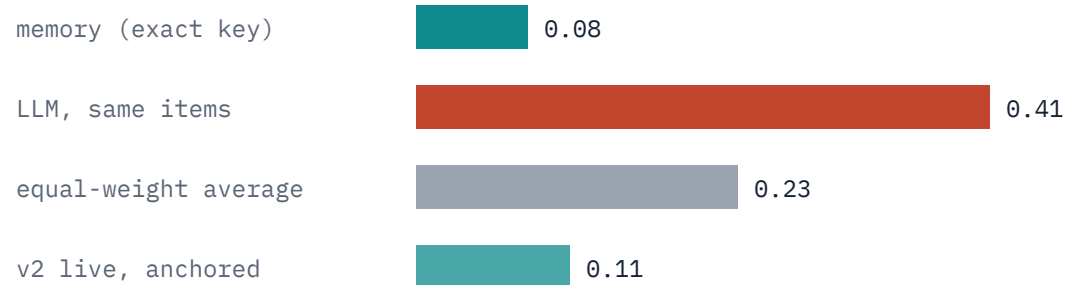
Income from truth brackets, expense against the charges actually billed to us. Shipped board to new pricing, our proven-fraud count unchanged.

ROUND	SHIPPED	NEW
41	–1,405	+49,042
42	–17,816	–17,400
45	+4,120	+18,239

The deviation metric alone would have picked a higher cap, because it costs under-charging only. Both sides, or nothing.

The market remembers what it paid. We made that the prior.

Median `|log error|` vs proven `t*`, same 26 items (rounds 38 to 45)



3-word keys: 68 % of items · 60 % of proven value carry a memory anchor

- Precision-weighted, so memory dominates where it exists (σ 0.15 vs 0.6); the LLM carries novel items and whales.
- Anchors go *inline per item* in the prompt, for example "paid ~EUR 58 to 86 per line when covered (n=11)", not one generic history block.
- Indemnity lines (stolen goods, declared values) take memory only as a floor: another case's watch says nothing about this one.

The v2 prompt, in full spirit

For every line: `p_covered` (0 to 1, calibrated: $p \geq 0.9$: 42 covered, 3 not; $p < 0.1$: 0 covered, 6 not), honest `q10` / `q50` / `q90`, a 12-word reason. Policy limits bind. Whales first. 1,400 characters; three parallel samples; first to land is submitted as insurance, the per-item median overwrites it.

207

memory keys after
game 96

15 s

typical first board on
the server

v2 live: three firsts in six rounds, and one round that taught us the most

ROUND	INCOME	EXPENSE	NET	RANK	NOTE
48	52,331	35,884	+16,447	1/17	27 items, first board 22.6 s
49	23,204	14,410	+8,794	1/17	
50	23,892	17,047	+6,845	4/17	
51	45,211	58,084	-12,873	10/17	two novel items, no memory
52	971	4,467	-3,496	15/17	2-item round
53	138,024	55,985	+82,039	1/17	whale charged at the proven max
54	n/a	n/a	-48,994		memory poisoning (slide 9)

Backtest, rounds 38 to 45 (same simulator, real opponents)

	EXPECTED	PESSIMISTIC
v1	162,489	-18,872
v2	170,576	-15,124
eyay (best team, actual)	175,917	n/a

Where memory covers most items (40, 42, 45) v2 is +24k / +20k / +6k. The two whale rounds carry 80 % of either total and hinge on the unproven upper side of t.

Every bad round produced a measurement, not an opinion

Truth from public payouts

Each round's feed proves brackets: a charge paid by all is fair ($t \geq a$); a charge refused by all and paid 0 is fraud ($t < a$). Matchup cells tighten them. 47 to 96 truth files.

Calibration with a drift alarm

Interval-censored MLE of bias and σ on current-prompt estimates only; a `live_shift` fitted on live rounds; make `learn` prints the measured $P(t \geq Q(1/3))$ against the $2/3$ the rule needs.

Replay, score, decide

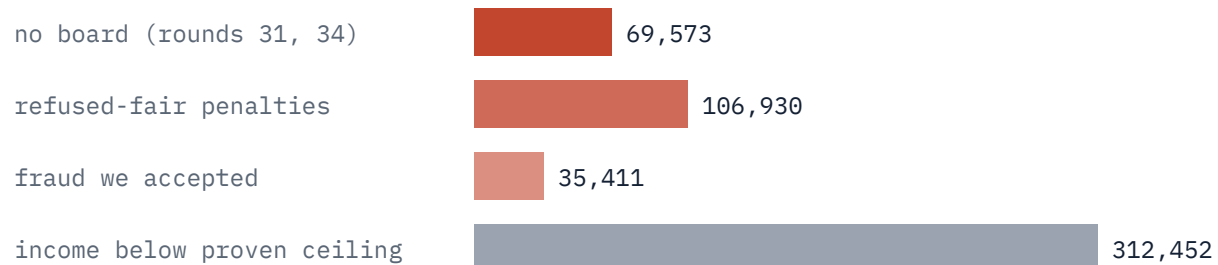
Stored estimates are re-priced in seconds; a change ships when it wins on money in a majority of rounds and on the two-sided counterfactual, never on a one-sided metric.

What the loop caught

- The belief's reliability *flipped sign* between prompt eras: claimed $P(t \geq b)$ 0.60, measured [0.67, 0.76] on the old prompt and [0.43, 0.43] live on the new one. Seven knobs were patching the same residual on different windows.
- Widening σ to the measured residual (≈ 1.15) *loses* money (−80k leave-one-out): the decision-relevant σ is not the residual σ , because the payoff is asymmetric and bounded.
- The "accept everything" idea: a team with a 93 % accept rate broke even and paid 25k in fraud in one round. The winning teams accept the right charges.

Where the money actually went (rounds 30 to 44: -165k)

EUR lost, by cause



Operational losses (two blank rounds, a Windows path / 7z chain, a 33 s call against a 60 s window) exceeded every pricing error of the same window.

The one that taught us the most: round 54, -48.6k

Game 7's air-con unit was *refused as not covered* ($t < 80.83$). v2's memory read that ceiling as a **price band of EUR 47 to 77**; the prompt made memory binding; the model obeyed: q50 = 62 on a EUR 2,000 unit, copied onto its sibling. b = 54 then refused every fair charge on the two biggest items.

Fix (v4): memory carries price from *paid* observations only; refusals never lower coverage; anchors advisory and quantity-aware. Same estimates, math only: **+17 %**; fresh replay turned -48.6k into -10.7k on that round.

Why we succeeded, and why we did not

Succeeded

- The measured-payoff maths paid immediately and measurably: step objective, derived b, coverage inside the rule: 31 % less regret re-priced, +49k / +18k on the counterfactual rounds.
- Market memory was the single best idea: 4× the model's precision on recurring lines, three firsts in v2's first six rounds, at a fifth of the latency.
- The learning loop was real (truth, calibration, reliability alarm, replay), so fixes were validated before they went live, and mistakes produced numbers.

Did not

- Operations cost more than modelling: ~120k to missed rounds and toolchain breakage; autoplay came late.
- The LLM's level error on first-sight items (+0.36, MAD 0.8 live) is the residual we never fixed. No constant can fix it, only more memory coverage or a prior for novel items.
- We tuned on windows that moved under us (replay vs live, old vs new prompt) several times before building the live-only monitor.
- Several engines in one working tree across machines: artefact collisions on every pull, and two machines once played the same round.